2nd UNIC-CASS Progress Questionnaire

The UNIC-CASS team asks each Design Team (DT) to fill out a short questionnaire, that will take about 5 minutes to complete.

Please answer all questions and write suggestions at the end. This will be useful for feedback by Mentors and Design to Tapeout Mentoring team to the students.

IEEE Circuits and Systems Society (CASS) UNIC-CASS page:
https://unic-cass.github.io/

vazquezleonardodavid@outlook.com Cambiar cuenta

No compartido

* Indica que la pregunta es obligatoria

Design Team Identifier (Use an ACRONYM as a short name for the IP block top level) *

QDLL

Primary contact email *

vazquezleonardodavid@outlook.com

Secondary contact email *

leonardo.vazquez@ieee.org

Active Design Team members (First Name & Last Name of all participants, except mentor) *

Leonardo David Vazquez
Santiago Basignana
Mateo Buteler
Clemente Molinari

Mentor's Name(s) *

Agustín Galetto

Repository link (e.g. https://github.com/username/repository-name) *

https://github.com/leonardovazquez/unic-cass-wrapper_QDLL

Link to tapeout report (provide link to e.g. Google docs file with your steps and outputs) *

https://docs.google.com/presentation/d/1SaUPbj3oBROP9z1-2TtK8tWIxL02-J9qEyfox9xo-iI/edit?usp=sharing

Design Team Country *

ARGENTINA

Is your design? *

- ○ Digital Only
- ◉ Analog / Mixed Signal

Do you have special I/O requirements (RF Signal, High Voltage, High Current, low R/C Parasitics, low leakage)?
If so, please explain with as much details as possible

The QDLL project basically consist on a single block with 1 differential input signal (2 inputs) and 1 differential output signal (2 outputs). For a CMOS (0-1.2 V) 250•MHz input signal, standard IO cells have been tested and found to perform well. However, if available, we prefer to interface with RF 50•Ω systems using IO cells with matched impedance. We would need the new RF iocells models. If this is not possible, as we have demonstrated, we will use the IO Cells sg13g2_IOPADAnalog that we mentioned in our document. Also, we can migrate our project to a lower frequency, there is no problem. Thanks!

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For Analog / Mixed-Signal Design

Is the circuit schematic for your Design already completed in Xschem? *

- (●) Yes
- ( ) No - Explain below

If the schematic is not completed, tell us where you are at

Tu respuesta

What is your Core Area in mm² (without the pads!) *

- (●) < 0.1 mm²
- ( ) 0.1 mm² to 0.2 mm²
- ( ) 0.2 mm² to 0.3 mm²
- ( ) 0.3 mm² to 0.4 mm²
- ( ) 0.4 mm² to 0.5 mm²
- ( ) 0.6 mm² to 0.7 mm²
- ( ) 0.7 mm² to 0.8 mm²
- ( ) 0.8 mm² to 0.9 mm²
- ( ) 0.9 mm² to 1 mm²
- ( ) > 1 mm²

How many VDD Pads (0-2) *

1

Which voltage option are you using as your supply voltage (VDD): *

- [x] 1.2V
- [ ] 3.3V

What is the expected current requirement for you supply pads (e.g., 0.1mA)

The average current on typical operation is 0.45 mA.
We preferred to use the 4mA supply pads

How many VSS Pads (0-2) *

1

How many IOVDD Pads (0-2) *

1

How many IOVSS Pads (0-2) *

1

How many Analog Pads (0-18) *

4

Have you completed the layout of what percentage of your full custom Design (choose from 0 = 0% to 10 = 100%) *

	1	2	3	4	5	6	7	8	9	10	
Nothing	○	○	○	○	○	○	○	○	○	○	Complete Layout

The digital blocks in your design, if needed, are they completed in RTL?

- ○ Yes
- ○ No
- ◉ Don't apply (fully analog)

Borrar la selección

What parts or blocks of your full custom Design are already laid-out and DRC & LVS verified with Klayout?

All blocks and the TOP have passed DRC and LVS verification with KLayout

Write two sentences about the status of your Design and the design team familiarity with the Analog Open-source EDA flow.

Most members of the design team are graduate and undergraduate students with less than one year of experience using analog open-source EDA tools. Nevertheless, we have become well familiarized with the design flow, supported by the fact that two team members completed an open-source analog design course taught by Prof. Jorge Marín within the UNICCAS program, which has been very beneficial for transferring knowledge to the rest of the team.

Write two sentences about difficulties, if any, the design team encountered with the Open-source tools and difficulties, PDK usage.

We encountered issues related to tool versions, as the UNICCAS repository does not provide the latest releases of Magic and KLayout. However, we have been able to work around these limitations and continue making progress in our project.

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